

AARHUS UNIVERSITY

COMPUTER SCIENCE

NUMERICAL LINEAR ALGEBRA

Handin 10

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I'm gonna be flipping burgers in 1 year ;-;

Anyway, consider the matrix

$$A = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.3 & 0.45 \\ 0.25 & 0.45 & 0.3 \end{bmatrix} \quad (1)$$

(a) Verify that A is symmetric and explain why A is diagonalizable with only real eigenvalues

Verifying that A is symmetric can be done by looking at it and saying 'Yep that's symmetric', since $row_i == column_i$

Or verify it using python.

```
1 A = np.array([[0.5, 0.25, 0.25],
2               [0.25, 0.3, 0.45],
3               [0.25, 0.45, 0.3]])
4
5 if np.array_equal(A, A.T):
6     print(True)
7 # Output for reference
8 True
```

Per the spectral theorem 23.10 [1] any real symmetric matrix $A \in \mathbb{R}^{n \times n}$ there exist an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ such that $A = V\Lambda V^T$ where Λ is a diagonal matrix, and because we can factor the matrix (1) into that, then it can be said to be diagonalizable.

To go even further, the Theorem also states that A has n real eigenvalues and an orthonormal basis consisting of it's eigenvectors...Yeah take that 7-year old Gauss! I'm smart too!

(b) Verify that 1.0 is an eigenvalue of A , and then determine the other eigen-values of A .

To verify it's possible to use proposition 24.4 [1] which says if A is a stochastic matrix, then 1.0 is an eigenvalue of A . A stochastic matrix is a square matrix $n \times n$ with non-negative entries and the sum of each row is 1.0. Looking at the matrix (1) it's trivial that the sum of each row is 1.0, so 1.0 is an eigenvalue of A .

To find the rest of the eigenvalues needed to construct the basis for A , the characteristic

polynomial for A is used:

$$\begin{aligned} p(\lambda) &= \det(A - \lambda I_3) = \det \begin{bmatrix} 0.5 - \lambda & 0.25 & 0.25 \\ 0.25 & 0.3 - \lambda & 0.45 \\ 0.25 & 0.45 & 0.3 - \lambda \end{bmatrix} \\ &= (0.5 - \lambda) \det \begin{bmatrix} 0.3 - \lambda & 0.45 \\ 0.45 & 0.3 - \lambda \end{bmatrix} - (0.25) \det \begin{bmatrix} 0 & 0.15 - \lambda \\ 0.25 & 0.3 - \lambda \end{bmatrix} + (0.25) \det \begin{bmatrix} 0.25 & 0.3 - \lambda \\ 0.25 & 0.45 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} p(\lambda) &= \det(A - \lambda I_3) = \det \begin{bmatrix} 0.5 - \lambda & 0.25 & 0.25 \\ 0.25 & 0.3 - \lambda & 0.45 \\ 0.25 & 0.45 & 0.3 - \lambda \end{bmatrix} \\ &= (0.5 - \lambda) \det \begin{bmatrix} 0.3 - \lambda & 0.45 \\ 0.45 & 0.3 - \lambda \end{bmatrix} - 0.25 \det \begin{bmatrix} 0.25 & 0.45 \\ 0.25 & 0.3 - \lambda \end{bmatrix} + 0.25 \det \begin{bmatrix} 0.25 & 0.3 - \lambda \\ 0.25 & 0.45 \end{bmatrix} \\ &= (0.5 - \lambda) ((0.3 - \lambda)^2 - 0.45^2) - 0.25 (0.25(0.3 - \lambda) - 0.25(0.45)) + 0.25 (0.25(0.45) - 0.25(0.3 - \lambda)) \\ &= (0.5 - \lambda) (\lambda^2 - 0.6\lambda - 0.1125) - 0.0625(-0.15 - \lambda) + 0.0625(0.15 + \lambda) \\ &= (-\lambda^3 + 1.1\lambda^2 - 0.1875\lambda - 0.05625) + 0.125\lambda + 0.01875 \\ &= -\lambda^3 + 1.1\lambda^2 - 0.0625\lambda - 0.0375 = p(\lambda) \end{aligned}$$

That was a lot of math...But it's not done yet, to extract the remaining eigenvalues a polynomial of the second degree is needed, not this wet dirty cloth of a third degree polynomial. BUT with the power of 23.1 [1], by knowing $\lambda_0 = 1$ the polynomial can be factored.

$$-\lambda^3 + 1.1\lambda^2 - 0.0625\lambda - 0.0375 = (\lambda - 1)q(\lambda)$$

where $q(\lambda)$ is polynomial of $n - 1$ degrees, which can be found by coefficient comparison.

$$-\lambda^3 + 1.1\lambda^2 - 0.0625\lambda - 0.0375 = (\lambda - 1)(A\lambda^2 + B\lambda + C) = A\lambda^3 + (B - A)\lambda^2 + (C - B)\lambda - C$$

It's known that $A = -1$ which is then used to determine B and C .

$$\begin{aligned} A &= -1 \\ B - A &= 1.1 \implies B = 0.1 \\ C - B &= -0.0625 \implies C = 0.0375 \\ -C &= -0.0375 \end{aligned}$$

Then inserting the values of A , B and C into $q(\lambda)$.

$$-\lambda^3 + 1.1\lambda^2 - 0.0625\lambda - 0.0375 = (\lambda - 1)(-\lambda^2 + 0.1\lambda + 0.0375) = 0$$

Now, since $\lambda_0 = 1$, is only possible if $(\lambda - 1) = 0$. this leaves the second term to be solved, which luckily for us is a second degree polynomial.

$$-\lambda^2 + 0.1\lambda + 0.0375 = 0$$

Quadratic formula go!

$$\lambda = \frac{-0.1 \pm \sqrt{(0.1)^2 - 4(-1)(0.0375)}}{2(-1)} = \frac{-0.1 \pm 0.4}{-2}$$

$$\lambda_1 = \frac{-0.1 + 0.4}{-2} = -0.15$$

$$\lambda_2 = \frac{-0.1 - 0.4}{-2} = 0.25$$

Wich give the eigenvalues of A as $\lambda_0 = 1$, $\lambda_1 = -0.15$ and $\lambda_2 = 0.25$.

(c) Compute an orthonormal basis for $\mathbb{R}^3$ consisting of eigenvectors of A .

The eigenvectors of A can be found by solving the system $(A - \lambda I)v = 0$ for each eigenvalue λ and a non-zero vector $v \in \mathbb{R}^3$

Note that whenever a free variable is found, the variable is set to 1

$$\lambda_0 = 1$$

$$(A - (1)I)v_0 = \begin{bmatrix} -0.5 & 0.25 & 0.25 \\ 0.25 & -0.7 & 0.45 \\ 0.25 & 0.45 & -0.7 \end{bmatrix} v_0 = 0$$

Continue with row reduction to find a free variable.

$$R_1 \rightarrow R_1 + \frac{1}{2}R_0$$

$$R_2 \rightarrow R_2 + \frac{1}{2}R_0$$

$$R_2 \rightarrow R_2 + R_1$$

$$\begin{bmatrix} -0.5 & 0.25 & 0.25 \\ 0 & -0.575 & 0.575 \\ 0 & 0 & 0 \end{bmatrix}$$

Back-substitution yields the eigenvector v_0

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_1 = -0.15$$

$$(A + (0.15)I_3)v_1 = \begin{bmatrix} 0.65 & 0.25 & 0.25 \\ 0.25 & 0.45 & 0.45 \\ 0.25 & 0.45 & 0.45 \end{bmatrix} v_1 = 0$$

Continue with row reduction to find a free variable.

$$R_2 \rightarrow R_2 - R_1$$

$$R_0 \rightarrow \frac{R_0}{2.6}$$

$$R_1 \rightarrow R_1 - R_0$$

$$\begin{bmatrix} 0.25 & 0.096 & 0.096 \\ 0 & 0.35 & 0.35 \\ 0 & 0 & 0 \end{bmatrix}$$

Back-substitution yields the eigenvector v_1

$$\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 0.25$$

$$(A - (0.25)I_3)v_2 = \begin{bmatrix} 0.25 & 0.25 & 0.25 \\ 0.25 & 0.05 & 0.45 \\ 0.25 & 0.45 & 0.05 \end{bmatrix} v_2 = 0$$

Continue with row reduction to find a free variable.

$$R_0 \rightarrow R_0 \cdot 2$$

$$R_0 \rightarrow R_0 - R_1$$

$$R_0 \rightarrow R_0 - R_2$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -0.4 & 0.4 \\ 0.25 & 0.45 & 0.05 \end{bmatrix}$$

Back-substitution yields the eigenvector v_2

$$\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

So A has the eigenvalues $\lambda \in \{1, -0.15, 0.25\}$ with the corresponding eigenvectors

$v_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $v_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ which, according the spectral theorem 23.10 [1] is an orthonormal basis for A on the form.

$$V_{fake} = \begin{bmatrix} 1 & 0 & -2 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

BUT WAIT Mr.Author! you're forgetting something! You're dumb!

This isn't orthonormal, you need to normalize the eigenvectors! Please don't call me dumb, I'm just a student trying to learn!

nah, you deserve that you poser, you starbucks sipping long haired hipster looking vinyl licking plant bingo wannabe. #SelfBurn

Anyway, the eigenvectors are normalised by dividing them with their respective norms.

$$\begin{aligned}v_0 &= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \\v_1 &= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\v_2 &= \frac{1}{\sqrt{6}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}\end{aligned}$$

So the real treasure (V) is

$$V = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \quad (2)$$

(d) Determine an orthogonal matrix V and a diagonal matrix Λ such that

$$A = V\Lambda V^T \quad (3)$$

V is determined in equation (2) and Λ is the diagonal matrix consisting of the eigenvalues

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.15 & 0 \\ 0 & 0 & 0.25 \end{bmatrix}$$

This can be verified with python to see if the determined matrices are correct.

```
1 V0 = np.array([1, 1, 1]) / np.sqrt(3)
2 V1 = np.array([0, -1, 1]) / np.sqrt(2)
3 V2 = np.array([-2, 1, 1]) / np.sqrt(6)
4
5 V = np.column_stack((V0, V1, V2))
6
7 Lambda = np.array([[1,0,0],
8                    [0,-0.15,0],
9                    [0,0,0.25]])
10 print(V@Lambda@V.T)
11
```

```

12 # Output for reference
13 [[0.5  0.25 0.25]
14  [0.25 0.3  0.45]
15  [0.25 0.45 0.3  ]]

```

Crazy! it works! I can't believe it, I'm so good at this! I'm a genius! I might be flipping burgers in 1 year!

(e) For

$$v = \begin{bmatrix} 0.12 \\ 0.21 \\ 0.39 \end{bmatrix} \quad (4)$$

Use the results to determine the limits of $A^k v$ for $k \rightarrow \infty$ Usage of the decomposition for A has been permitted by a higher power, (The Authors Brain).

$$A^k v = (V \Lambda V^T)^k v = V \Lambda^k V^T v$$

$$\Lambda^k = \begin{bmatrix} 1^k & 0 & 0 \\ 0 & (-0.15)^k & 0 \\ 0 & 0 & (0.25)^k \end{bmatrix}$$

Letting $k \rightarrow \infty$ means the values -0.15^∞ and 0.25^∞ get the big ol' zero treatment which leaves 1^∞ , and that is always just 1.

So the matrix Λ becomes

$$\Lambda^{k \rightarrow \infty} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

So when Λ gets multiplied by V , only the first column remains

$$V \Lambda^{k \rightarrow \infty} = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & 0 \\ \frac{1}{\sqrt{3}} & 0 & 0 \\ \frac{1}{\sqrt{3}} & 0 & 0 \end{bmatrix}$$

and then $(V \Lambda^{k \rightarrow \infty}) V^T$

$$(V \Lambda^{k \rightarrow \infty}) V^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & 0 \\ \frac{1}{\sqrt{3}} & 0 & 0 \\ \frac{1}{\sqrt{3}} & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

finally multiplying the result of $V \Lambda^\infty V^T$ with v

$$(V \Lambda^{k \rightarrow \infty} V^T) v = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0.12 \\ 0.21 \\ 0.39 \end{bmatrix} = \begin{bmatrix} 0.24 \\ 0.24 \\ 0.24 \end{bmatrix}$$

So the limit of $A^k v$ for $k \rightarrow \infty$ is $\begin{bmatrix} 0.24 \\ 0.24 \\ 0.24 \end{bmatrix}$

Appendix

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 A = np.array([[0.5, 0.25, 0.25],
5               [0.25, 0.3, 0.45],
6               [0.25, 0.45, 0.3]])
7
8 if np.array_equal(A, A.T):
9     print(True)
10
11 V0 = np.array([1, 1, 1]) / np.sqrt(3)
12 V1 = np.array([0, -1, 1]) / np.sqrt(2)
13 V2 = np.array([-2, 1, 1]) / np.sqrt(6)
14
15 V = np.column_stack((V0, V1, V2))
16
17 Lambda = np.diag([1, -0.15, 0.25])
18 print(V@Lambda@V.T)
```

References

- [1] Andrew Swan. *Numerisk Linær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.